P3 - Modules, Arrays and Attractor Architecture

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Approach: Parametric

Revision: 2026-10-19. Current assignment requirements and deadlines. Completed midterm work is reviewed using the evidence already submitted; no retrospective deliverables are added.

3.1 - Descriptive geometry - build a reusable module

Introduced: Week 4 - Sep 15 / **Due:** Sun Oct 18, 11:59 PM Eastern Time / **Weight:** 10%

Purpose

Build a panel from a documented real-world example by describing its geometry, then turn it into a reusable module with meaningful inputs.

What you will learn

- Understand points/vertices, lines/edges, faces and meshes, and how they connect.
- Understand a variable, a list and an index; make point and face lists and reference their entries.
- Understand a parameter as a control and the module as a first function: inputs produce geometry.

What to do

- Choose and credit a real panel or architectural system. Show enough reference material to explain your interpretation.
- Construct the panel using explicit points and faces that refer to those points. Show readable point and face lists; no separate explanation of face winding or normals is required.
- Expose a few meaningful parameters, such as width, opening or depth. Show three settings that reveal how the same module changes.
- Use one material and consistent three-point lighting for the comparison. The lighting tutorial is in Resources; no lighting diagram or material-settings table is required.
- Check that the module works when repeated. For a complex panel with 24 or more points, a Boolean may supplement the construction if at least 16 explicit points generate geometry; review this route with your professor.

What to submit

- **Presentation:** one 17 x 11 inch landscape sheet, `Lastname_Firstname_P3a.pdf`.
- **On the sheet:** a credited panel precedent; three rendered variations of the same module; a point list with readable indices and coordinates; a face list referencing those point indices; and a small table showing the parameter values for all three variations.
- **Comparison:** label the three variations and show which controls changed. Use the same material, view and lighting setup. No lighting diagram, material-settings table or separate normals explanation.
- **Editable model:** `.blend`, or `.gh` plus referenced `.3dm`, with the module inputs and geometry construction accessible. Package textures/assets; keep the selected module ready for reuse in 3.2 and 3.5.
- **Package:** `Lastname_Firstname_P3a/`. Check that changing the named inputs reproduces the three shown variations and that the point/face lists describe the submitted module.

What we assess

- Can you explain how the lists construct the mesh, how an index selects an entry and how a parameter changes the module? Are the variations and documentation clear?
- Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

3.2 - Arrays - repetition, patterns and architecture

Introduced: Week 6 - Sep 29 / **Due:** Sun Oct 18, 11:59 PM Eastern Time / **Weight:** 10%

Purpose

Reuse your module to learn repetition, make conditional patterns and develop a small architectural application. Parts 3.2.1-3.2.3 form one assignment.

What you will learn

- 3.2.1: understand for loops, other repetition methods, nested repetition and the role of the index.
- 3.2.2: understand if/Switch and conditions. Modulo returns the remainder after division and can select repeating groups, such as every second or third element. Use arithmetic on an index to vary a parameter.
- 3.2.3: combine your module, repetition and pattern rules into an architectural object.

What to do

- 3.2.1 - Repetition: show the module alone, then in a 1D row, 2D grid and 3D array. Label counts and spacing. Identify how repetition is nested and what each index refers to.
- 3.2.2 - Conditional patterns: show four patterns on a consistent array. Include a modulo-based rule and an index-driven parameter change. Caption each with a simple rule such as "shift every other row" or "increase the opening along the row."
- 3.2.3 - Architectural application: combine the previous studies into one simple facade, wall, canopy, building or similar object. Show an overall view and a supporting view explaining the module and pattern.
- Reuse your module and one readable graphic setup. Describe the rules in ordinary language; formal equations, count products and polished lighting variations are not required.

What to submit

- **Presentation:** exactly three 17 x 11 inch landscape sheets in one file, `Lastname_Firstname_P3b.pdf`. Parts 3.2.1-3.2.3 are one submission and one grade.
- **Sheet 1 - 3.2.1:** four diagrams: module alone, 1D row, 2D grid and 3D array. Label counts, spacing and direction; identify the nested repetition and the role of each index.
- **Sheet 2 - 3.2.2:** four conditional patterns on a consistent array. Label each pattern with its rule and the parameter it changes. Identify the modulo rule and the index-driven variation.
- **Sheet 3 - 3.2.3:** one architectural application with an overall view and a supporting view explaining how the module and pattern form a facade, wall, canopy, building or similar object. State the intended use in a short caption.
- **Editable files:** `.blend`, or `.gh` plus referenced `.3dm`, with the three studies clearly identified and dependencies packaged in `Lastname_Firstname_P3b/`.
- **Check:** controls reproduce the diagrams and every sheet uses the GSM. Completed midterm array studies remain valid evidence; this sheet structure does not require retrospective resubmission.

What we assess

- Explain the repeated unit, nesting, index and condition; demonstrate that changing a control changes the intended elements; show a coherent architectural application.
- One 10% assignment, not three separate grades. Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

3.4 - Attractors - three field studies

Introduced: Week 9 - Oct 20 / **Due:** Week 10 - Oct 27, 11:59 PM Eastern Time / **Weight:** 0%

Purpose

Compare how different attractors produce a varying pattern across the same module facade. This is preparation for 3.5, not an additional graded project.

What you will learn

- Understand a field as values evaluated across locations; an attribute can store values on geometry.
- Use distance to a point, several points or a curve to drive a module parameter.
- Adjust the influence range and map it to a useful design range.

What to do

- Use the same facade, module, view and controlled parameter for all three studies.
- Diagram 1: one point attractor. Diagram 2: multiple point attractors. Diagram 3: one curve attractor.
- Show the attractors and the resulting module variation. Label the parameter and describe how distance changes it.
- For multiple points, state how their influences are combined, such as the nearest point or an average. Show how moving an attractor changes the pattern; no formal equation is required.

What to submit

- **Presentation:** one 17 x 11 inch landscape study sheet. It becomes page 1 of the combined `Lastname_Firstname_P3c.pdf` submitted with 3.5 on October 27; no separate upload or grade.
- **Three diagrams:** the same module facade with (1) one point attractor, (2) multiple point attractors and (3) a curve attractor. Use the same view and controlled panel parameter for a clear comparison.
- **Labels:** show the attractor locations/curve, name the panel parameter being varied and give a short explanation of the distance response. For multiple points, state how the influences combine.
- **Editable files:** retain all three studies in the combined 3.4-3.5 source package. Check that moving each attractor changes the intended module geometry.

What we assess

- Can you explain the field, locate its controls and predict the effect of moving an attractor? Are the three methods clearly distinguishable?
- Formative feedback; this sheet supports the 3.5 assessment.

3.5 - Attractor-based architectural application

Introduced: Week 9 - Oct 20 / **Due:** Week 10 - Oct 27, 11:59 PM Eastern Time / **Weight:** 10%

Purpose

Use your existing panel and attractor logic to design something architectural. Develop a simple facade, wall, canopy, pavilion or similar application with a clear purpose and scale.

What you will learn

- Develop an architectural application from your existing panel, connecting the repeated module to an overall form, spatial purpose and human scale.
- Use attractor-driven variation for an architectural purpose, such as opening size, depth, orientation or enclosure.
- Refine a simple project through controlled changes and clear drawings.

What to do

- Reuse the panel you developed in 3.1 to design a simple facade, wall, canopy, pavilion or similar architectural application. You may develop your 3.2 architectural study further.
- Give the design a clear purpose and scale. Explain what the panel system encloses, screens, shades or defines, and how attractor-driven variation supports that purpose.
- Choose point, multiple-point or curve attractors. You do not need to combine every attractor type in the final design.
- Show one controlled comparison and a resolved elevation and isometric. Identify the attractor, the geometric parameter it drives and why the variation matters architecturally.
- A new precedent study is not required. Facadepot and other references may provide optional inspiration; credit any images or ideas you use.

What to submit

- **Presentation:** two 17 x 11 inch landscape sheets together in `Lastname_Firstname_P3c.pdf`, due October 27. Page 1 is the existing 3.4 study sheet; page 2 is the 3.5 architectural application.
- **Page 2 drawings:** identify the reused 3.1 panel; show a resolved elevation and isometric of the same architectural design; include a small before/after comparison showing the effect of the attractor.
- **Page 2 labels:** indicate the attractor and the parameter it controls, explain the architectural purpose in a short caption, and show scale with a person and key dimensions. A new precedent study is not required.
- **Editable model:** `.blend`, or `.gh` plus referenced `.3dm`, containing the three 3.4 studies and the final architectural application. Include textures and other referenced assets.
- **Package:** `Lastname_Firstname_P3c/`. Check that the elevation and isometric agree, the existing panel is reused and the visible attractors produce the documented variation. Two sheets total; reuse the study sheet.

What we assess

- Clear architectural application, an understandable attractor system, purposeful variation and coherent presentation.
- The combined 3.4-3.5 submission carries 10%. Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

Shared standards: Apply the GSM and credit sources. Package linked assets; check PDFs and editable files before uploading. Explain rules plainly and follow the course submission policies.